What Mathematical Work Do AI Tutors Leave for Learners? A Qualitative Study and a Test of Decision-Guided Tutoring

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Abstract: AI tutors must help students without completing the mathematical reasoning they need to practice. Using TutorMoments-Preview, a dataset of real tutoring conversations, we examine this problem in two ways. First, we code the work left for learners in 38 moments. Explicit instructions about balancing support and challenge produced more replies inviting connections, explanations, or strategies than general tutoring guidance. Second, we test whether a separate model can improve an open-weight tutor by choosing scaffolding or rigor before the tutor replies. We compare Jev and OpenAI Decisions with a tutor choosing its own approach and with fixed instructions across 520 moments. Automated first-reply pass rates were 68.7% for self-choice, 72.9% with Jev, 73.5% with Decisions, and 74.0% with fixed rigor guidance. The results show modest gains over self-choice but no advantage over the strongest simple control, highlighting the distinction between selecting a teaching direction and delivering appropriate support.

Introduction

A mathematics tutor must decide when to help and when to leave the next step to the learner. *Scaffolding* makes a task accessible, for example by offering a hint or a useful representation. *Pushing for rigor* asks the learner to do more demanding mathematical work, such as explaining a relationship or choosing a strategy. Too much help can remove the reasoning that the student needs to practice. Too little can leave the student stuck. This assistance dilemma is central to the design of tutoring systems (Koedinger & Aleven, 2007). Research on teacher questioning also shows why the work elicited from students matters, beyond whether the teacher's explanation is correct (Franke et al., 2009).

We study this problem using TutorMoments-Preview, a public collection of real tutoring conversations and AI-generated continuations (Zhang et al., 2026). Our project has two connected parts. First, we examine what mathematical work remains for a learner after a tutor responds. Second, we test a technical approach intended to improve that response: a separate model reads the conversation, chooses scaffolding or rigor, and selects instructions for the model that writes the tutor's reply. The idea is that a small tutor might respond more appropriately if it receives a clear teaching decision before generating text.

The approach builds on work that separates instructional planning from language generation. StratL guides tutors through teaching intents (Puech et al., 2025), while DICE separates decisions about whether to intervene from the ability to produce an answer (Pal et al., 2026). Eduardo takes a different approach by training smaller tutors through reinforcement learning (Macina et al., 2026). We investigate whether an additional decision step can help an existing model without retraining it. Our questions are: **(1)** How do tutoring instructions change the mathematical work left for learners? **(2)** Does a separate scaffold-or-rigor decision improve a smaller model's reply beyond detailed instructions alone, and what does that decision step add in cost and delay?

What TutorMoments contains and how we use it

TutorMoments-Preview contains 462 de-identified transcripts of remote tutoring with students in grades 2–7. A transcript records dialogue and descriptions of shared-screen activity. A *moment* is a stopping point in a conversation where a tutor must decide what to do next. Experienced teachers described important situations, the human tutor's action, and what followed. The dataset authors used language models to map these descriptions to categories and combined annotations by majority vote. Their balanced evaluation set contains 520 moments: 260 labeled as opportunities for scaffolding and 260 as opportunities to push for rigor (Zhang et al., 2026).

For each moment, the benchmark gives an AI tutor the conversation up to that point and asks it to continue with a simulated learner. It compares two kinds of instructions. The *plain prompt* gives general tutoring guidance. The *evaluation-aware prompt* explains scaffolding, rigor, and over-scaffolding, and asks the tutor to choose appropriately. It does not reveal the moment's assigned category. The authors found that explicit guidance improved performance, although models still tended to give too much help and miss opportunities for greater challenge. Their scoring can credit a response that combines scaffolding and rigor, provided it avoids over-scaffolding. The categories therefore identify instructional opportunities, not two mutually exclusive forms of good teaching.

Our qualitative study uses the released human and AI continuations. Our technical study generates new tutor replies from the same conversation histories. Neither study collects new classroom data. We examine the first response after each stopping point, rather than reproducing the benchmark's multi-turn exchange. Our percentages consequently measure a different outcome from its published leaderboard.

Study 1: What work does the tutor leave for the learner?

We compared the human continuation with GPT-5.5 continuations under the plain and evaluation-aware prompts at 38 mathematics moments, one per student and 19 per source category. The sample began with 40 moments selected from 226 eligible situations after excluding students used in earlier pilots. Two were removed because intervening learner activity made the human response an unsuitable comparison. They were not replaced. A separate audit established a 346-moment mathematics comparison set with clear conversation boundaries and usable responses across the original conditions. The supplement records the complete selection process and exclusions.

The unit of analysis is the first tutor response block, combining consecutive tutor utterances until the learner next speaks or acts. Following a theory-informed content-analysis approach (Hsieh & Shannon, 2005), we developed two codes. *Work Left* describes the most demanding mathematical contribution still available to the learner: no new work or repetition (W0), a specified calculation or heavily cued recognition (W1), a connection or explanation within a narrowed path (W2), or a strategy, argument, or diagnosis with meaningful discretion (W3). *Information Supplied* describes how much of the current answer the tutor provides, from none (R0), through general guidance (R1) and a task-specific step or relationship (R2), to the complete answer (R3). Unclear cases receive NA. A higher W code does not necessarily mean better teaching: the appropriate demand depends on the learner.

Two AI-assisted coding passes applied the codebook to responses with their source identities hidden and preceding dialogue available. We report the first pass and retain the second to show how interpretation changes the results. All 114 response texts and their supporting quotations were checked against the source packets. The main comparison is between 38 pairs of AI replies. Human continuations provide a naturalistic reference because they were not produced under controlled prompt conditions.

Qualitative findings

Evaluation-aware replies more often left a connection, explanation, or strategy for the learner: 16/38 received W2 or W3, compared with 2/38 plain replies (Table 1). Plain replies supplied the complete active answer in 21/38 cases, compared with 5/38 evaluation-aware replies. One plain reply was unclear on this measure.

Table 1: Mathematical work left in the first coding pass ($n = 38$ per column).

Work left	Plain	Evaluation-aware	Human reference
W0: No new work or repetition	21	2	20
W1: Calculation or cued recognition	15	20	12
W2: Connection or explanation	2	14	1
W3: Strategy, argument, or diagnosis	0	2	0
NA: Unclear demand	0	0	5

For example, one problem asked the learner to complete $25 \times 14 = 25 \times (2 \times \star)$. The plain reply stated, "So the star should be 7," then verified the equality. The evaluation-aware reply asked, "What number times 2 makes 14?" The latter still supplied the decisive relationship, but left the calculation to the student. This distinction is easy to miss if analysis only records whether the tutor asked a question. The second coding pass also favored evaluation-aware replies, assigning W2/W3 to 23 rather than five plain replies. Its different counts show that boundaries between calculation and mathematical explanation require judgment.

Study 2: Can a separate decision improve the next tutor reply?

The proposed approach

We attempted a technical contribution by separating two tasks normally handled by one language model: deciding what kind of support is needed and writing that support. A *decision model*, also called a router here, reads the conversation and returns either scaffolding or rigor. That choice selects one of two detailed instruction sets for the tutor. Scaffolding instructions ask for the smallest useful hint while leaving the key calculation or inference to the student. Rigor instructions ask the tutor to invite a justification, connection, or independent step without supplying the reasoning first. In both cases, the tutor still receives the full conversation and must write a reply appropriate to

the current problem.

We tested two decision systems: Jev 1.14.0, a TypeSafe model designed to make fast structured choices (Almeida, 2026), and OpenAI's Decisions API using GPT-6 Luna (OpenAI, n.d.). Qwen3-32B, an open-weight language model, wrote the tutor replies (Qwen Team, 2025). The intended sequence is therefore: *conversation so far → scaffold-or-rigor decision → corresponding instructions → Qwen tutor reply*. The motivation is practical: specialized decision making could improve a less expensive tutor, but it is only useful if the resulting replies justify the extra step.

Separating decision accuracy from response quality

We first tested whether each decision system matched the dataset's assigned category across all 520 moments. The selected prompts produced 376 matches for Jev (72.3%) and 386 for Decisions (74.2%). These predictions were saved and reused in the response experiment. Prompt development had already used this dataset, so these are exploratory results on previously inspected cases. A subsequent prompt revision reduced agreement for both systems. The supplement preserves all versions.

We then tested what those choices meant for the actual tutor reply. At every moment, Qwen generated three alternatives: **self-choice**, where it received both detailed instruction sets and chose an approach itself; **scaffold-guided**, where it received the scaffolding instructions; and **rigor-guided**, where it received the rigor instructions. All three also received the publisher's evaluation-aware prompt. This produced 1,560 replies. For each saved Jev or Decisions prediction, we selected the corresponding scaffold-guided or rigor-guided reply. Generating both alternatives let us compare the decision systems on exactly the same candidate text, without confounding their choices with different generations.

We included two simple controls: always use scaffold guidance and always use rigor guidance. These test whether a separate decision model adds value beyond a consistent instruction. A further diagnostic selects the reply corresponding to the dataset label, showing what happens when that category is supplied correctly. This label is never provided to either decision model. All models receive the complete preceding dialogue, with future turns and teacher annotations excluded. The tutor additionally receives the publisher's student metadata. Generation used temperature zero, non-thinking mode, and a 384-token output limit. No reply was truncated.

How replies were evaluated

Gemini 2.5 Flash evaluated the three replies in randomized order, without condition names or dataset labels. A reply passed when it displayed the behavior called for by the dataset label, avoided over-helping and mathematical errors, and fit the conversation. A reply judged to combine scaffolding and rigor could satisfy either category. We also asked Claude Sonnet 4.6 to evaluate the same replies for a fixed subset of 32 students, balanced by category. Comparisons pair replies from the same moments, with uncertainty intervals obtained by resampling students so that related moments remain together.

Technical findings

Jev-guided replies passed in 379/520 cases and Decisions-guided replies in 382, compared with 357 for self-choice (Table 2). These were gains of 4.2 percentage points for Jev (95% student-resampling interval: -1.0 to 9.2) and 4.8 points for Decisions (0.2 to 9.5). However, always using rigor guidance passed 385 cases, slightly more than either decision system. Jev improved performance on rigor opportunities from 149 to 180 passes, while reducing scaffolding passes from 208 to 199.

Table 2: Passing first replies. Each row uses the same 520 conversation stopping points.

How instructions are selected	All 520	Scaffold 260	Rigor 260	Math set 346
Qwen chooses its approach	68.7%	80.0%	57.3%	72.0%
Jev chooses	72.9%	76.5%	69.2%	72.8%
OpenAI Decisions chooses	73.5%	80.8%	66.2%	75.1%
Always use scaffold instructions	56.5%	86.2%	26.9%	59.8%
Always use rigor instructions	74.0%	65.8%	82.3%	76.0%
Use the dataset label (diagnostic)	84.2%	86.2%	82.3%	86.4%

The mathematics set also requires clear turn boundaries and usable original responses; it is not a subject-only filter. The diagnostic assumes access to the assigned label and is not a deployable system. These are our first-reply ratings, not the original multi-turn benchmark score.

In the stricter 346-moment mathematics set, Jev's advantage over self-choice was only 3 cases, or 0.9

points (interval -5.3 to 6.5). Evaluation was also sensitive to the judge. On the same 32 cases, Gemini passed 20 self-choice and 19 Jev replies; Claude passed 16 and 22, respectively. The two evaluators agreed on 75 of the 96 individual reply judgments (78.1%).

Correct category selection did not guarantee a good response. Of Jev's 376 category matches, 320 selected replies passed; 59 of its 144 mismatches also passed. In one case, both decision systems correctly selected scaffolding for a learner working on $1{,}000 - 10$, but Qwen's reply returned to an earlier $338 + 586$ problem. Even selecting by the dataset label passed only 438/520 cases. The remaining failures concern how the tutor carries out its instruction, not just which instruction is chosen.

We found no speed advantage. Qwen's median time to begin answering was 1.20 seconds for self-choice. Adding the separately measured decision durations gave estimated totals of 1.43 seconds with Jev and 1.60 with Decisions. These are reconstructed estimates, because saved decisions selected pre-generated replies; we did not time a live end-to-end system. Estimated model costs per 1,000 replies were \$0.59, \$0.90, and \$1.25, respectively, excluding evaluation.

Discussion and contribution

The qualitative analysis describes what a learner can contribute after a reply. The technical analysis tests whether a separate decision helps produce that reply. Together they show why choosing an instructional category is only part of the tutoring problem. A model can choose scaffolding correctly yet supply too much information or address the wrong task. Conversely, a response can combine support and challenge even when its assigned instruction names only one of them.

Our experiment does not establish that adding Jev or Decisions is better than giving Qwen a clear, fixed teaching instruction. It does identify an essential comparison for future systems: a decision step should beat both equally detailed self-choice and simple fixed guidance. The fixed-rigor result should not be read as advice to demand more difficulty from every student. The evaluator identified both scaffolding and rigor in 202 of its 520 replies. Its instruction could therefore encourage independent reasoning while the generated response still offered support. A useful next decision target may specify the learner's next mathematical action and what information to withhold, rather than only selecting a broad category.

These findings come from selected benchmark moments, previously inspected data, AI-assisted coding, and automated judgments. They do not measure student learning. The human replies are observations from real sessions, while the new AI replies were not delivered to learners. Independent human evaluation and testing on new conversations are needed before drawing conclusions about classroom benefit. The contribution of this course study is a concrete coding account of mathematical work and a reproducible, controlled test showing where decision-guided tutoring helps, where simple guidance performs similarly, and why category agreement alone is insufficient evidence of a better tutor.

Supplementary materials

The qualitative analysis folder provides the codebook, coded responses, results, and source conversations. The technical analysis folder provides the full methods, prompts, predictions, replies, ratings, and runnable code. A GitHub repository maintains the implementation; the Drive copy is accessible without repository access. AI tools assisted coding, programming, evaluation, and drafting. No independent human coding validation is claimed.

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